

Heart Disease Prediction Using Machine Learning

This report presents a complete machine learning workflow for predicting heart disease using medical and lifestyle patient data.

1. Data Preprocessing

Preprocessing steps included duplicate removal, handling missing values, outlier removal using IQR, log transformation, encoding categorical variables, and feature scaling.

1.1 Removing Unnecessary Columns

The id column was removed because it has no predictive value.

1.2 Duplicate Removal

Duplicate rows were removed to avoid biased learning.

1.3 Missing Values Handling

Missing numerical values were filled using the mean.

Categorical missing values were filled using the mode.

1.4 Outlier Detection and Removal

Outliers were detected using the IQR method.

$$IQR = Q_3 - Q_1$$

Values outside the acceptable range were removed to reduce noise and improve model performance.

1.5 Log Transformation

The ST depression feature showed a right-skewed distribution.

To reduce skewness, log transformation was applied:

$$ST_{log} = \log (1 + ST \text{ depression})$$

This transformation reduced the effect of extreme values and improved feature distribution.

1.6 Encoding Categorical Variables

Categorical variables were converted into numerical values using One-Hot Encoding.

1.7 Feature Scaling

Standardization was applied to normalize feature values.

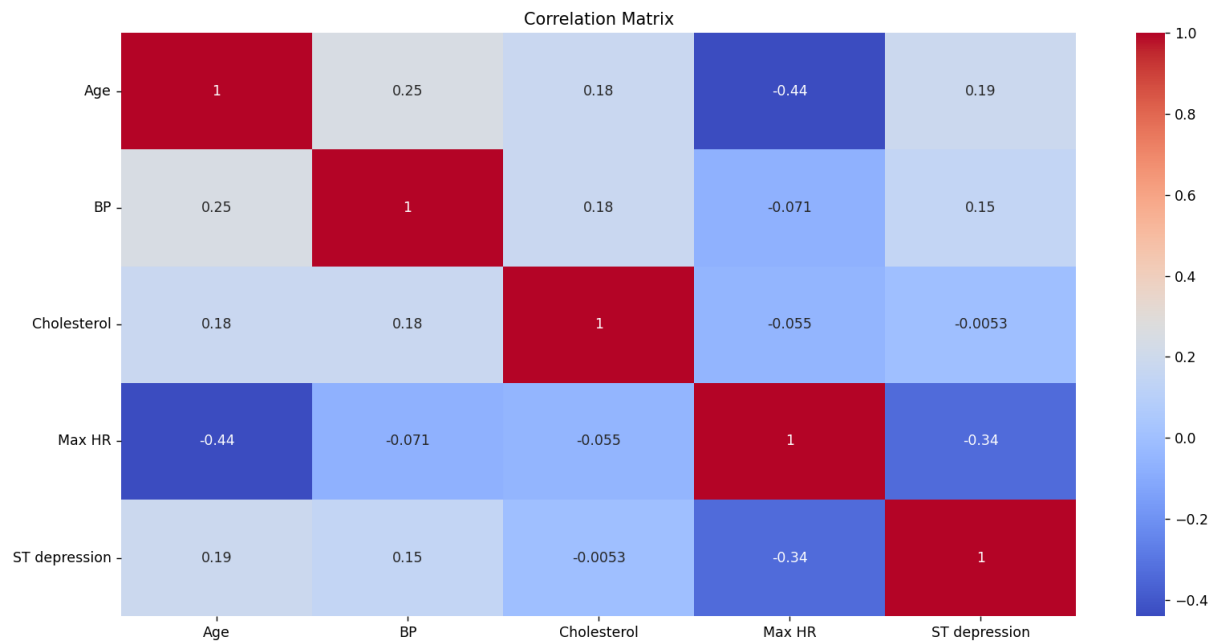
This step was especially important for:

- Logistic Regression
- KNN
- SVM

2.Feature Engineering

Several engineered features were created such as HR_diff, Age_group, High_Risk, and stress_ratio to improve model performance

Correlation Heatmap



Feature	Correlation
---------	-------------

Thallium	0.54
----------	------

Number of vessels fluro	0.45
-------------------------	------

ST depression	0.40
---------------	------

Chest pain type	0.39
-----------------	------

Exercise angina	0.38
-----------------	------

HR_diff	0.35
---------	------

Max_HR	-0.39
--------	-------

2.1 Age Group Feature

Patient ages were grouped into categories:

Group Age Range

0	0–25
1	25–50
2	50–70
3	70–100

This simplified age-related cardiovascular risk representation.

2.2 Heart Rate Difference Feature (HR_diff)

A new feature was created based on the medical formula:

$$\text{Max Heart Rate} \approx 220 - \text{Age}$$

The difference between expected and actual heart rate was calculated:

$$HR_{diff} = (220 - \text{Age}) - \text{MaxHR}$$

This feature captures abnormal heart rate behavior.

2.3 High Risk Feature

A binary feature indicating patients with both:

- High blood pressure
- High cholesterol

Conditions:

- $BP > 140 \& \text{Cholesterol} > 240$

This feature identifies patients with elevated cardiovascular risk.

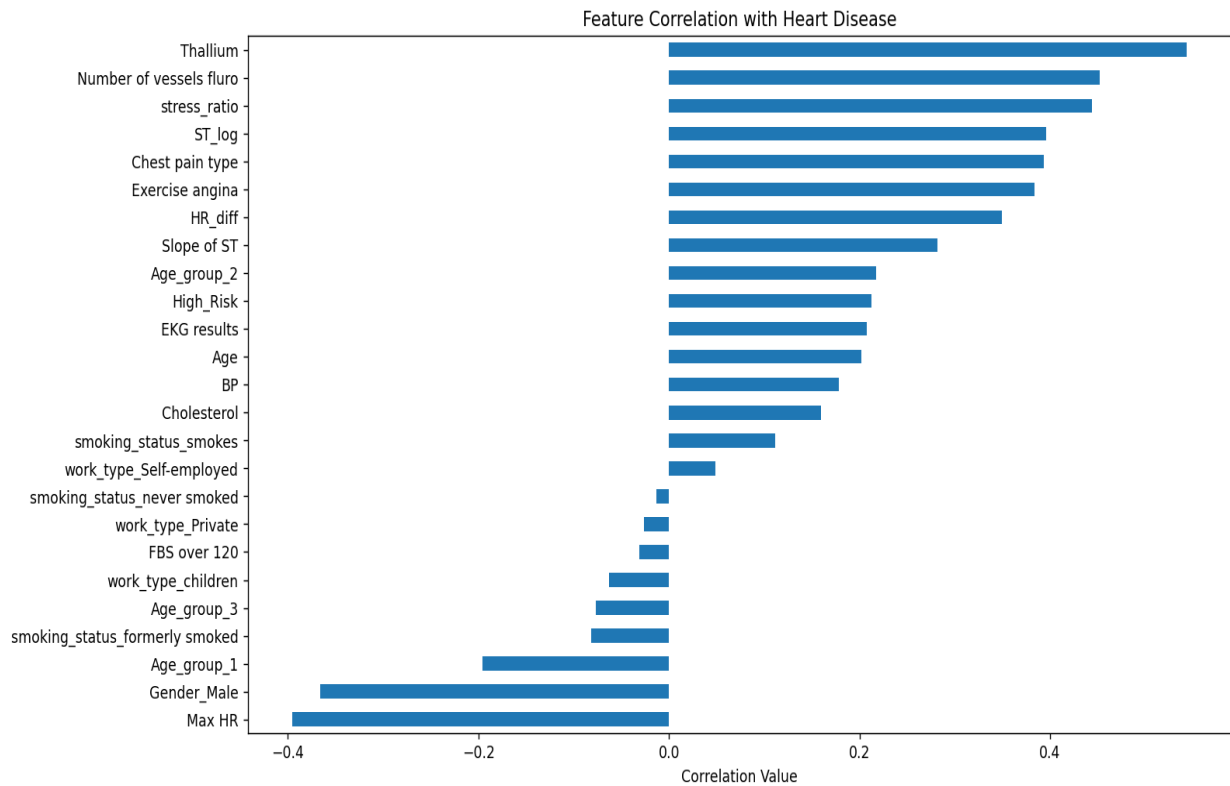
2.4 Stress Ratio Feature

A stress-related feature was engineered using:

$$\text{Stress Ratio} = \frac{ST_{log}}{MaxHR}$$

This combines ECG stress measurements with heart rate performance.

Feature correlation with target



3. Machine Learning Models

Four classification models were trained and evaluated.

Models Used

1. Logistic Regression
2. Decision Tree
3. K-Nearest Neighbors (KNN)
4. Support Vector Machine (SVM)

4. Hyperparameter Tuning

Several experiments were performed to optimize model performance.

Logistic Regression C Experiments

- Small C values apply stronger regularization.
- Medium C values balance learning and generalization.
- Very large C values may cause overfitting.

Decision Tree max_depth Experiments

- Very small depth values underfit the data.
- max_depth = 3 achieved the best balance between complexity and generalization.
- Larger depth values caused overfitting and reduced validation accuracy.

KNN n_neighbors Experiments

- Smaller K values make the model sensitive to noise.
- Medium K values balance generalization.
- Larger K values create smoother decision boundaries.

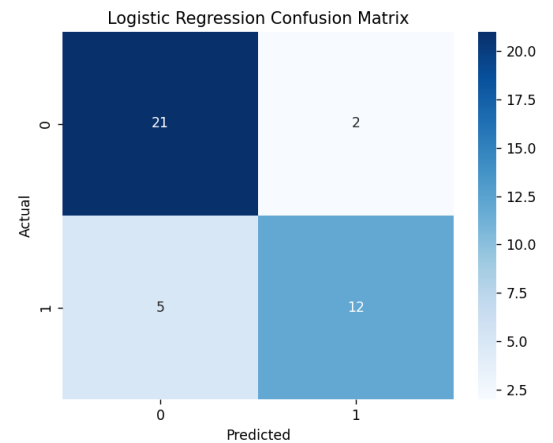
SVM C Experiments

- Small C values allow simpler decision boundaries.
- - Larger C values try harder to classify all training samples correctly.
- - Very large C values may cause overfitting.

Logistic Regression Analysis

Confusion Matrix

The Logistic Regression model achieved an accuracy of 82.5%, demonstrating solid overall classification performance. The confusion matrix showed that the model correctly classified 21 healthy patients (True Negatives) and 12 heart disease patients (True Positives). However, 5 patients with heart disease were incorrectly classified as healthy (False Negatives), while 2 healthy patients were incorrectly predicted as diseased (False Positives).



Logistic Regression C Experiments

C = 0.01 -> Accuracy = 0.8000

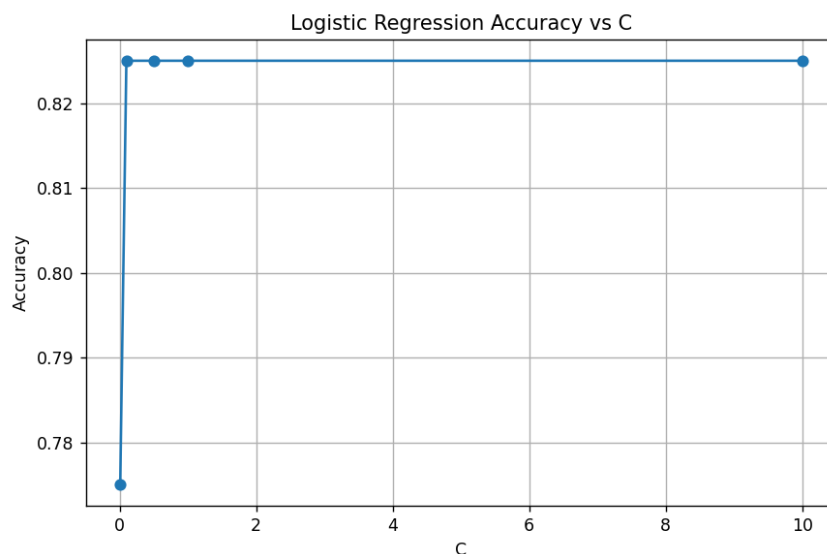
C = 0.1 -> Accuracy = 0.8250

C = 0.5 -> Accuracy = 0.8250

C = 1 -> Accuracy = 0.8250

C = 10 -> Accuracy = 0.8250

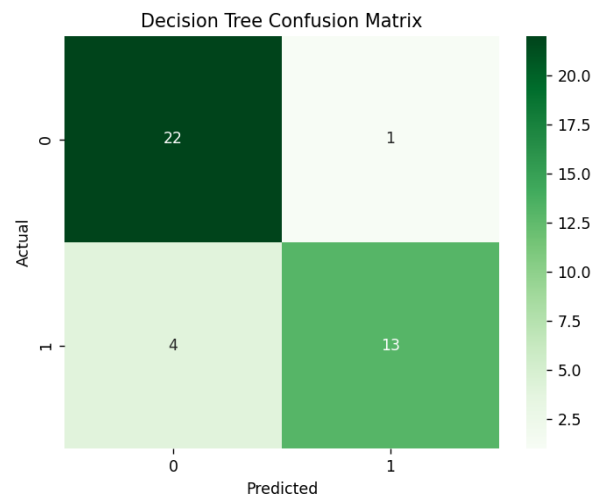
Different C values were tested.



Decision Tree Analysis

Confusion Matrix

The Decision Tree model achieved the best overall performance with an accuracy of 87.5%. The confusion matrix showed that the model correctly classified 22 healthy patients and 13 heart disease patients, while producing only 1 False Positive and 4 False Negatives.



Decision Tree max_depth Experiments

max_depth = 2 -> Accuracy = 0.7250

max_depth = 3 -> Accuracy = 0.8750

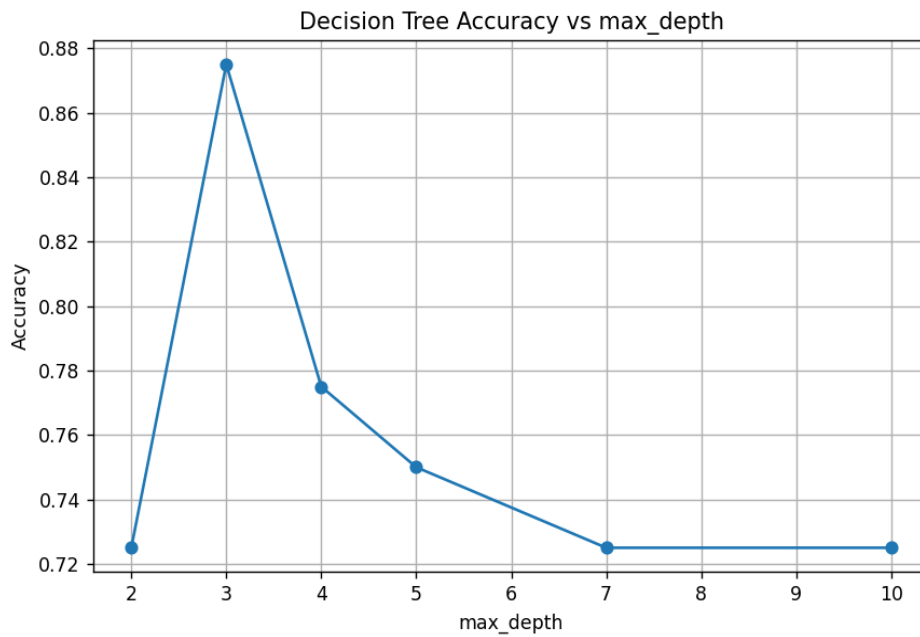
max_depth = 4 -> Accuracy = 0.7750

max_depth = 5 -> Accuracy = 0.7500

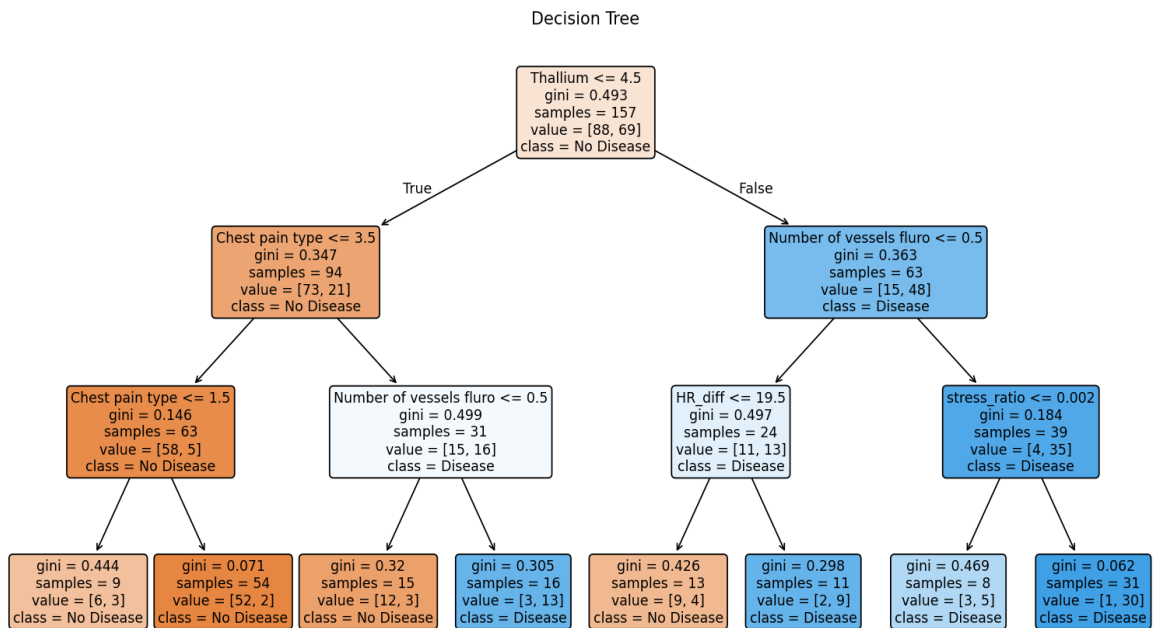
max_depth = 7 -> Accuracy = 0.7250

max_depth = 10 -> Accuracy = 0.7250

Different “max_depth” values were tested



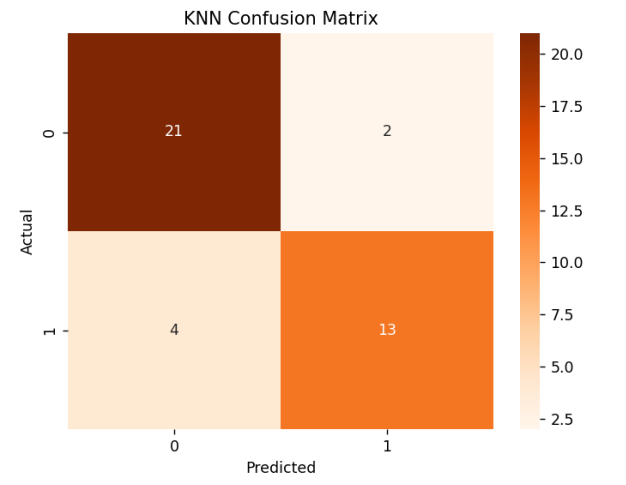
Tree



KNN Analysis

Confusion Matrix

The KNN model achieved an accuracy of 85%, showing competitive performance compared to the other models. The confusion matrix indicated that the model correctly classified 21 healthy patients and 13 diseased patients, while producing 2 False Positives and 4 False Negatives.



KNN n_neighbors Experiments

n_neighbors = 1 -> Accuracy = 0.7250

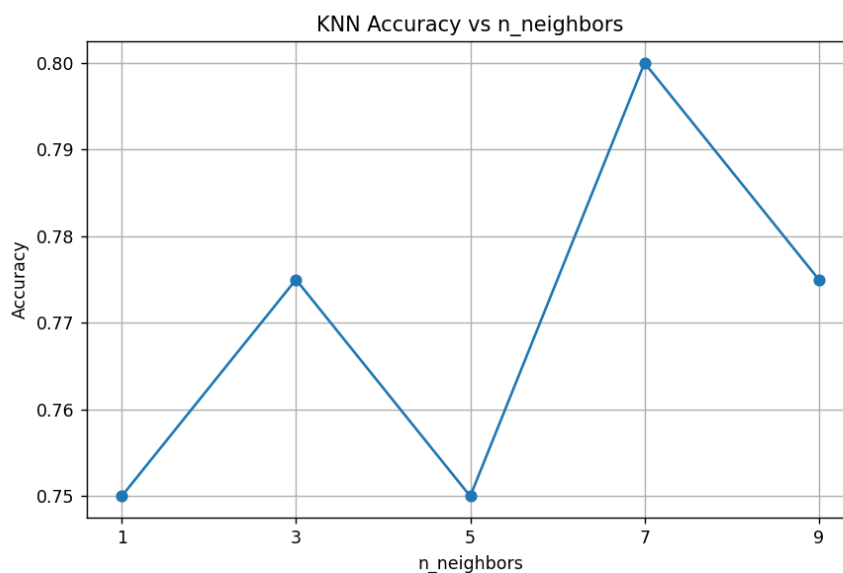
n_neighbors = 3 -> Accuracy = 0.8500

n_neighbors = 5 -> Accuracy = 0.7500

n_neighbors = 7 -> Accuracy = 0.8250

n_neighbors = 9 -> Accuracy = 0.7500

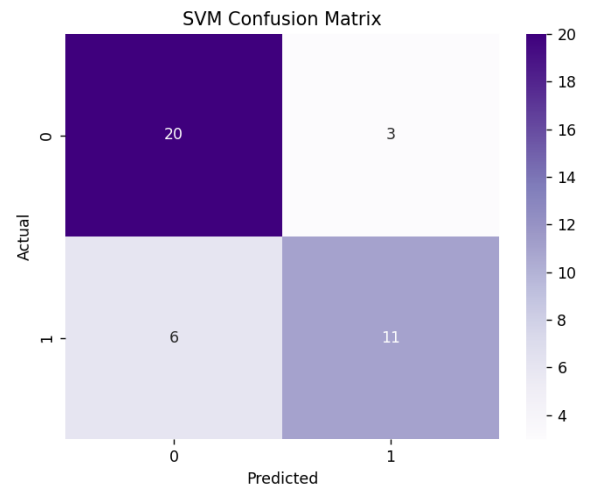
Different neighbor counts (k) were tested.



SVM Analysis

Confusion Matrix

The SVM model achieved an accuracy of 77.5%, which was lower than the Decision Tree and KNN models. The confusion matrix showed that the model correctly classified 20 healthy patients and 11 diseased patients, while producing 3 False Positives and 6 False Negatives.



SVM C Experiments

C = 0.01 -> Accuracy = 0.5750

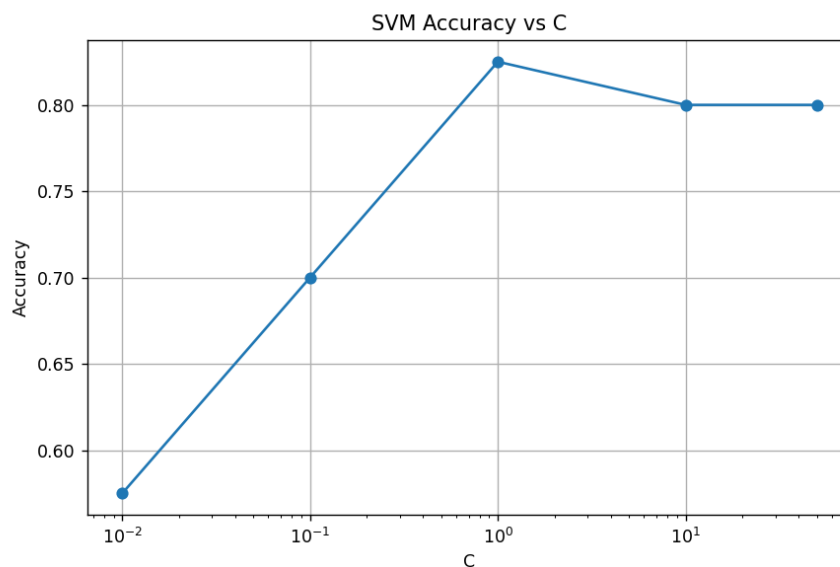
C = 0.1 -> Accuracy = 0.6750

C = 1 -> Accuracy = 0.8500

C = 10 -> Accuracy = 0.7750

C = 50 -> Accuracy = 0.7750

Different C values were tested



FINAL CONCLUSION

=====

This project successfully applied machine learning techniques to predict heart disease using patient medical data.

Several preprocessing techniques were applied including:

- handling missing values
- removing outliers
- feature engineering
- log transformation
- encoding categorical variables
- feature scaling

Four classification models were trained and evaluated:

- Logistic Regression

- Decision Tree

- KNN

- SVM

Hyperparameter experiments were also performed to study the impact of model complexity and hyperparameter tuning.

Decision Tree achieved the best overall performance because it balanced model simplicity and generalization effectively with accuracy 87.5%.

